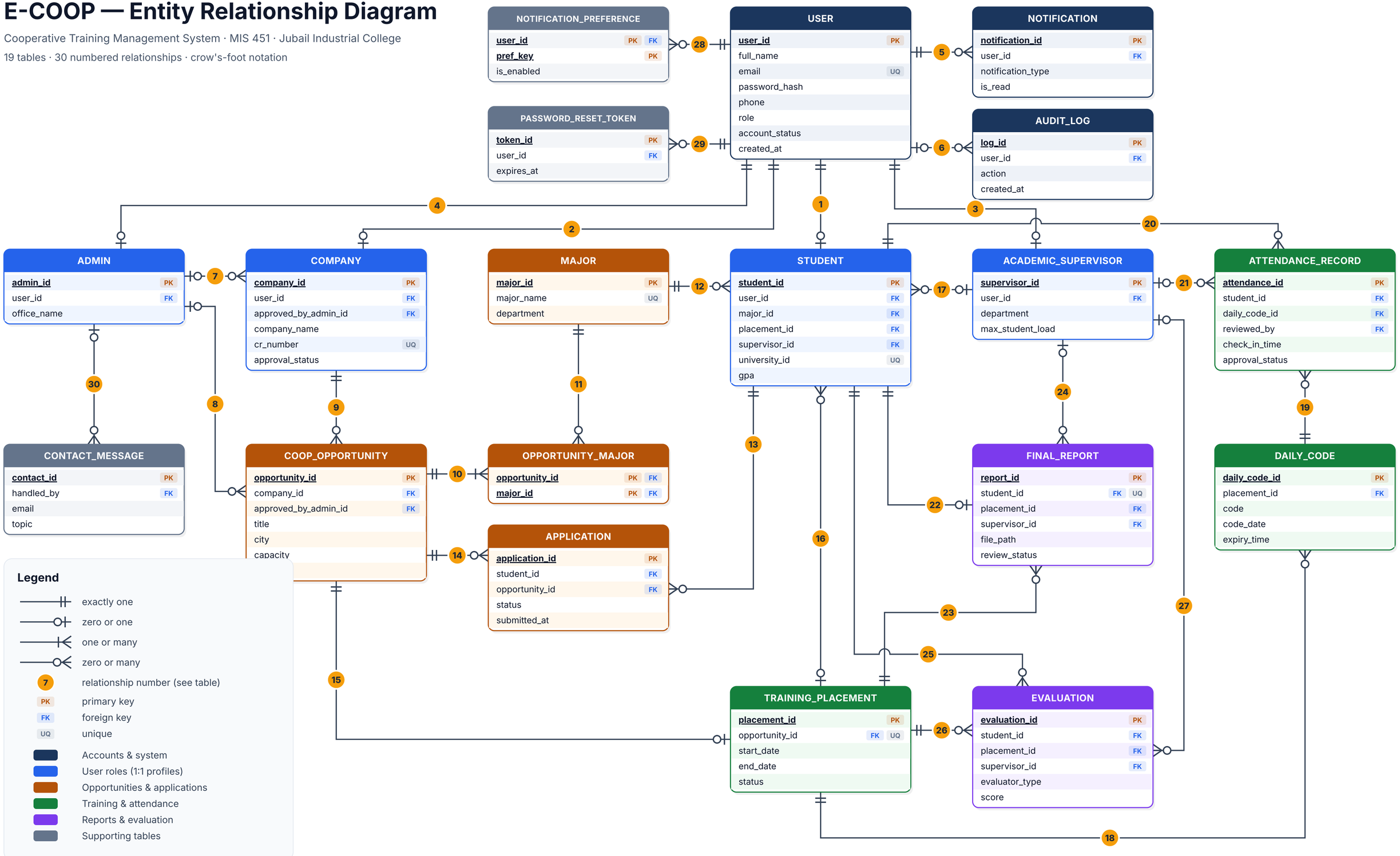
E-COOP — Entity Relationship Diagram

Cooperative Training Management System · MIS 451 · Jubail Industrial College

19 tables · 30 numbered relationships · crow's-foot notation



E-COOP — Numbered relationships

Read each row as: parent (cardinality) — (cardinality) child. Every relationship is one foreign key in the MySQL schema.

#	Relationship	Foreign key	Explanation (both directions)
1	**USER** 1 — 0..1 **STUDENT** has profile	`students.user_id`	→ Each USER may have zero or one STUDENT profile. ← Each STUDENT belongs to exactly one USER.
2	**USER** 1 — 0..1 **COMPANY** has profile	`companies.user_id`	→ Each USER may have zero or one COMPANY profile. ← Each COMPANY belongs to exactly one USER.
3	**USER** 1 — 0..1 **ACADEMIC_SUPERVISOR** has profile	`academic_supervisors.user_id`	→ Each USER may have zero or one ACADEMIC_SUPERVISOR profile. ← Each ACADEMIC_SUPERVISOR belongs to exactly one USER.
4	**USER** 1 — 0..1 **ADMIN** has profile	`admins.user_id`	→ Each USER may have zero or one ADMIN profile. ← Each ADMIN belongs to exactly one USER.
5	**USER** 1 — 0..N **NOTIFICATION** receives	`notifications.user_id`	→ Each USER receives zero or many NOTIFICATIONs. ← Each NOTIFICATION is sent to exactly one USER.
6	**USER** 0..1 — 0..N **AUDIT_LOG** generates	`audit_logs.user_id`	→ Each USER generates zero or many AUDIT_LOG entries. ← Each AUDIT_LOG entry is linked to zero or one USER (empty for a failed login with an unknown email).
7	**ADMIN** 0..1 — 0..N **COMPANY** approves	`companies.approved_by_admin_id`	→ Each ADMIN approves zero or many COMPANIES. ← Each COMPANY is approved by zero or one ADMIN (none while still pending).
8	**ADMIN** 0..1 — 0..N **COOP_OPPORTUNITY** approves	`coop_opportunities.approved_by_admin_id`	→ Each ADMIN approves zero or many COOP_OPPORTUNITIES. ← Each COOP_OPPORTUNITY is approved by zero or one ADMIN (none while still pending).
9	**COMPANY** 1 — 0..N **COOP_OPPORTUNITY** posts	`coop_opportunities.company_id`	→ Each COMPANY posts zero or many COOP_OPPORTUNITIES. ← Each COOP_OPPORTUNITY is posted by exactly one COMPANY.
10	**COOP_OPPORTUNITY** 1 — 1..N **OPPORTUNITY_MAJOR** requires	`opportunity_majors.opportunity_id`	→ Each COOP_OPPORTUNITY requires one or many OPPORTUNITY_MAJOR rows (at least one major). ← Each OPPORTUNITY_MAJOR row belongs to exactly one COOP_OPPORTUNITY.
11	**MAJOR** 1 — 0..N **OPPORTUNITY_MAJOR** required by	`opportunity_majors.major_id`	→ Each MAJOR appears in zero or many OPPORTUNITY_MAJOR rows. ← Each OPPORTUNITY_MAJOR row refers to exactly one MAJOR. (10 + 11 = the M:N between opportunities and majors.)
12	**MAJOR** 1 — 0..N **STUDENT** enrolls	`students.major_id`	→ Each MAJOR enrolls zero or many STUDENTs. ← Each STUDENT is enrolled in exactly one MAJOR.
13	**STUDENT** 1 — 0..N **APPLICATION** submits	`applications.student_id`	→ Each STUDENT submits zero or many APPLICATIONs (one per opportunity). ← Each APPLICATION is submitted by exactly one STUDENT.
14	**COOP_OPPORTUNITY** 1 — 0..N **APPLICATION** receives	`applications.opportunity_id`	→ Each COOP_OPPORTUNITY receives zero or many APPLICATIONs. ← Each APPLICATION is for exactly one COOP_OPPORTUNITY.
15	**COOP_OPPORTUNITY** 1 — 0..1 **TRAINING_PLACEMENT** results in	`training_placements.opportunity_id (UNIQUE)`	→ Each COOP_OPPORTUNITY results in zero or one TRAINING_PLACEMENT (BR-10). ← Each TRAINING_PLACEMENT belongs to exactly one COOP_OPPORTUNITY.
16	**TRAINING_PLACEMENT** 0..1 — 0..N **STUDENT** includes	`students.placement_id`	→ Each TRAINING_PLACEMENT includes zero or many STUDENTs. ← Each STUDENT is in zero or one TRAINING_PLACEMENT (none until accepted).
17	**ACADEMIC_SUPERVISOR** 0..1 — 0..N **STUDENT** supervises	`students.supervisor_id`	→ Each ACADEMIC_SUPERVISOR supervises zero or many STUDENTs (up to max_student_load). ← Each STUDENT is supervised by zero or one ACADEMIC_SUPERVISOR (none until assigned).
18	**TRAINING_PLACEMENT** 1 — 0..N **DAILY_CODE** has	`daily_codes.placement_id`	→ Each TRAINING_PLACEMENT has zero or many DAILY_CODEs (one per day). ← Each DAILY_CODE belongs to exactly one TRAINING_PLACEMENT.
19	**DAILY_CODE** 1 — 0..N **ATTENDANCE_RECORD** validates	`attendance_records.daily_code_id`	→ Each DAILY_CODE validates zero or many ATTENDANCE_RECORDs. ← Each ATTENDANCE_RECORD is validated by exactly one DAILY_CODE.
20	**STUDENT** 1 — 0..N **ATTENDANCE_RECORD** checks in	`attendance_records.student_id`	→ Each STUDENT checks in with zero or many ATTENDANCE_RECORDs. ← Each ATTENDANCE_RECORD belongs to exactly one STUDENT.

21	**ACADEMIC_SUPERVISOR** 0..1 — 0..N **ATTENDANCE_RECORD** approves	`attendance_records.reviewed_by`	→ Each ACADEMIC_SUPERVISOR approves zero or many ATTENDANCE_RECORDs. ← Each ATTENDANCE_RECORD is approved by zero or one ACADEMIC_SUPERVISOR (none while pending).
22	**STUDENT** 1 — 0..1 **FINAL_REPORT** submits	`final_reports.student_id (UNIQUE)`	→ Each STUDENT submits zero or one FINAL_REPORT. ← Each FINAL_REPORT is submitted by exactly one STUDENT.
23	**TRAINING_PLACEMENT** 1 — 0..N **FINAL_REPORT** has	`final_reports.placement_id`	→ Each TRAINING_PLACEMENT has zero or many FINAL_REPORTs. ← Each FINAL_REPORT belongs to exactly one TRAINING_PLACEMENT.
24	**ACADEMIC_SUPERVISOR** 0..1 — 0..N **FINAL_REPORT** reviews	`final_reports.supervisor_id`	→ Each ACADEMIC_SUPERVISOR reviews zero or many FINAL_REPORTs. ← Each FINAL_REPORT is reviewed by zero or one ACADEMIC_SUPERVISOR (none until reviewed).
25	**STUDENT** 1 — 0..N **EVALUATION** receives	`evaluations.student_id`	→ Each STUDENT receives zero or many EVALUATIONs (one from the company, one from the supervisor). ← Each EVALUATION is about exactly one STUDENT.
26	**TRAINING_PLACEMENT** 1 — 0..N **EVALUATION** has	`evaluations.placement_id`	→ Each TRAINING_PLACEMENT has zero or many EVALUATIONs. ← Each EVALUATION belongs to exactly one TRAINING_PLACEMENT (this gives the evaluating company).
27	**ACADEMIC_SUPERVISOR** 0..1 — 0..N **EVALUATION** evaluates	`evaluations.supervisor_id`	→ Each ACADEMIC_SUPERVISOR gives zero or many EVALUATIONs. ← Each EVALUATION is given by zero or one ACADEMIC_SUPERVISOR (empty when the company is the evaluator).
28	**USER** 1 — 0..N **NOTIFICATION_PREFERENCE** sets	`notification_preferences.user_id`	→ Each USER sets zero or many NOTIFICATION_PREFERENCEs. ← Each NOTIFICATION_PREFERENCE belongs to exactly one USER.
29	**USER** 1 — 0..N **PASSWORD_RESET_TOKEN** requests	`password_reset_tokens.user_id`	→ Each USER requests zero or many PASSWORD_RESET_TOKENs. ← Each PASSWORD_RESET_TOKEN belongs to exactly one USER.
30	**ADMIN** 0..1 — 0..N **CONTACT_MESSAGE** handles	`contact_messages.handled_by`	→ Each ADMIN handles zero or many CONTACT_MESSAGEs. ← Each CONTACT_MESSAGE is handled by zero or one ADMIN (none while new).